

Kartikay

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EXPERIENCE

• AI Research Engineer

January 2026 - Present

Metafore

Bengaluru

- Built an end-to-end model fine-tuning platform, streamlining the pipeline from dataset preparation through training to deployment for downstream teams.
- Engineered ontology-aware knowledge graph workflows integrated with LLM agents to structure and reason over domain data.
- Developed a UI-driven agent generator that scaffolds production agents directly from a Business Requirements Document (BRD).

• Software Engineering Consultant

August 2025 - Jan 2026

Assured Health Technologies

Bengaluru

- Created 17 data migration scripts in Python that ran across the database to update various fields.
- Used AWS for deployment and automated production operations using CloudFormation, decreasing deployment time by 67%.
- Optimized Django API to reduce time from 35.7s to 3.7s using indexes and removing extra mutational queries on prefetching.
- Utilized Bun runtime to reduce compile and run time by 53%.

• Software Engineering Intern under LitmusChaos

July 2024 - September 2024

Cloud Native Computing Foundation Intern

Remote

- Developed an upgrade agent streamlining version transitions for Litmus, leading to a 30% reduction in manual intervention time during upgrades and enhancing overall user experience across the platform.
- Engineered sequential upgrade pipeline reducing version migration time for control plane and database updates.
- Automated the process of upgrading the MongoDB schema with every new version of Litmus via manifests.
- Optimized the login API of the Auth server by 98.84% by bringing down the latency from 1.73s to 15.87ms.

• Google Summer Of Code (Intern at Organic Maps)

April 2023 - July 2023

Software Developer Intern

Remote

- Built high-performance proxy using TypeScript and NodeJS, serving 50K+ requests/day with 99.9% uptime on Cloudflare Workers.
- Architected proxy to convert Google Maps URIs to custom formats, reducing load time by 60%.
- Integrated the newly prepared proxy seamlessly into both iOS and Android applications, enabling a streamlined API interaction that improved data fetching speed by 30% across mobile platforms.

PERSONAL PROJECTS

• LLM Inference Engine

A from-scratch LLM inference engine in PyTorch (KV cache, batching, paged memory)

Github

- Built a from-scratch LLM inference engine in PyTorch with KV caching, continuous batching, and paged memory.
- Achieved 2.22x decode speedup from KV caching at 1K tokens.
- Gained 118x throughput via batched inference on A100; cut memory 8.7x with a paged cache.
- Hit 11.1x serving throughput under 64 concurrent users; validated across 122 tests.

• gogent

A Go framework for LLM agents with built-in OpenTelemetry observability

Github

- Designed a Go framework for building LLM agents, providing a structured runtime for tool use, provider integration, and agent orchestration.
- Built in OpenTelemetry observability to emit traces and metrics for agent runs, making multi-step agent pipelines debuggable and monitorable.
- Abstracted LLM provider access behind a common interface, supporting multiple providers and enabling drop-in swapping without changing agent code.
- Included a memory subsystem and examples suite to demonstrate reusable agent patterns out of the box.

TECHNICAL SKILLS AND INTERESTS

Languages: JavaScript, TypeScript, Go, Python, Rust, C++, PHP, SQL

Frameworks: ExpressJS, ReactJS, Socket.io, Web3JS, jQuery

Cloud/Databases: AWS, Ansible, Docker, Prometheus, Apache Kafka, Thanos, OpenTelemetry, Kubernetes, MongoDB, MySQL

Coursework: Udemy certifications of Web Development and Kubernetes, completed Google Cloud Engineer track for Google Cloud event, GDSC IIT Patna, MA101 and MA102 (Linear algebra and real analysis)

EDUCATION

•Indian Institute of Technology Patna
Bachelor of Technology

2025